

AWS Data Pipeline vs Glue vs Lambda: Who Is a Clear Winner?

[AWS](#) provides users with some of the most effective ETL tools for streamlined data management. Whether you are willing to implement a new platform, undertake third-party integrations, or simply move all your data to a warehouse, these ETL tools help you in managing your database in a secure and private manner.

However, it is important to select the right ETL tool for AWS depending on your specific requirements. Here, we would compare three of such tools – AWS Data Pipeline, AWS Glue, and AWS Lambda.

What Is AWS Data Pipeline?

AWS [Data Pipeline](#) is an ETL tool by Amazon that helps users automate data transfer processes. It helps you move data through dedicated and automated workflows that make data tasks dependent on the tasks completed successfully earlier.

The [AWS Data Pipeline workflows](#) allow users to automate and leverage their ETL processes on the AWS cloud, making them take an advantage of the features already existing on the platform. Moreover, the tool is suitable for both technical and non-technical users as it provides you with a simple drag-and-drop interface. This allows users to have complete control of specific computational resources pertaining to the Data Pipeline logic.

The aim of AWS Data Pipeline is to automate seamless data movement within the AWS cloud by defining, scheduling, and automating individual tasks. For example, if a user is willing to extract event data from a specific source on a regular basis, they can do so by designing a data pipeline and running the same on an Amazon EMR over the concerned datasets for generating extensive reports.

AWS Data Pipeline makes data management easier as it allows users to transfer and transform their datasets across different AWS tools and monitor all relevant processes from a centralized location.

Important Features of AWS Data Pipeline

Here are some of the most important features of AWS Data Pipeline:

- The tool makes it easy for users to debug or change the logic of your automated data workflow by providing complete control to compute resources required to execute business logic.
- The tool provides users with an architecture that is high on flexibility and tolerance. This allows Data Pipeline to run and monitor data processing activities with ease and efficiency.
- The flexible nature of the tool allows users to write their own conditions or use the pre-built conditions to make use of features such as error handling, scheduling transfers, etc.
- The tool provides users with seamless support for an array of data sources that range from AWS cloud to on-premise data sources.
- It allows users to define activities like HiveActivity, SQLActivity, PigActivity, EMRAActivity, and more for effective transformation of data on the AWS cloud.
- AWS Data Pipeline charges users \$1 for every pipeline for running it more than once a day and \$0.68 per pipeline per month if it is run once or lesser in a day.

What Is AWS Glue?

AWS Glue is a fully managed ETL tool by Amazon that provides users with quick and efficient ways of performing a range of activities like data enriching, data cleaning, data cleaning, and many more between data stores and streams.

The tool is designed to work with a semi-structured database and consists of three major components – Data Catalog, Scheduler, and ETL Engine. It also provides users with the feature of Dynamic Frame – a data abstraction feature that helps users organize their data into set rows and columns. Here, each record is self-describing and does not require the users to specify a schema.

AWS Glue is also a big data cataloging tool that helps users perform ETL processes on the AWS cloud. For example, users can create and run an ETL job in their AWS Management Console using the AWS Glue interface

and point AWS Glue to their data. This allows the tool to store specific metadata in the Data Catalog and generate code to execute data transformations and other relevant processes.

Important Features of AWS Glue

Here are some of the most important features of AWS Glue:

- The tool automatically generates code for performing ETL processes after users specify the location/path where the concerned data needs to be stored.
- The tool allows users to set up crawlers for connecting them to data sources. This helps them in classifying the datasets, obtaining schema, and storing the same in the data catalog automatically.
- Users set up continuous ingestion pipelines and prepare streaming data on the go using the serverless streaming ETL function.
- AWS provides users with an integrated data catalog with table definitions and other relevant control information for managing the AWS Glue environment.
- AWS Glue costs \$0.44 for every Data Processing Unit hour, billed with every second of the tool being used. Also, users are charged with \$1 per 100,000 objects managed in the data catalog and \$1 per million requests made to the data catalog.

What Is AWS Lambda?

AWS Lambda is a computing service that allows you to run code without the need for provisioning or managing servers. The tool runs code on high-availability computing infrastructure and allows users to perform complete administration of the compute resources. This includes processes like server and operating system maintenance, code monitoring and logging, and automatic scaling.

AWS Lambda allows users to run code for any kind of application or back-end service as required. All they need to do is supply code to a language supported by the tool.

Important Features of AWS Lambda

Here are some of the most important features of AWS Lambda:

- The tool allows users to add custom logic to specific AWS resources like Amazon S3 buckets and DynamoDB tables. This makes it easy for users to compute data as it moves through the AWS cloud.
- The tool allows users to create new back-end services for their applications triggered on-demand with the help of Lambda API or custom API developed with the help of Amazon API Gateway.
- AWS Lambda users do not need to learn new languages, frameworks, and tools. They can use any suitable third-party library, including the native ones.
- The tool relieves users from building dedicated back-end services by running code on a fault-tolerant infrastructure. Using Lambda there is no need to update the existing OS when a new patch is released also similarly when a resize or update is made to the servers as the usage increases.
- AWS Lambda charges users \$0.20 per million requests and \$0.0000166667 for every GB-second use of the tool.

Which Tool Is a Clear Winner?

Each of the AWS ETL tools has its own niche, purpose, and scale of usage. All of these tools provide you with ease of operation and process automation based on your specific requirements. It is advisable to assess your specific data management requirements and budget constraints before making a calculated choice.